

BOOST CONTROL CARD - FABRICATION DRAWING

Release candidate BOOST_control_RC1, 2026-09-24. Files: BOOST_control_RC1_gerbbers.zip. Units: mm.

FABRICATION NOTES

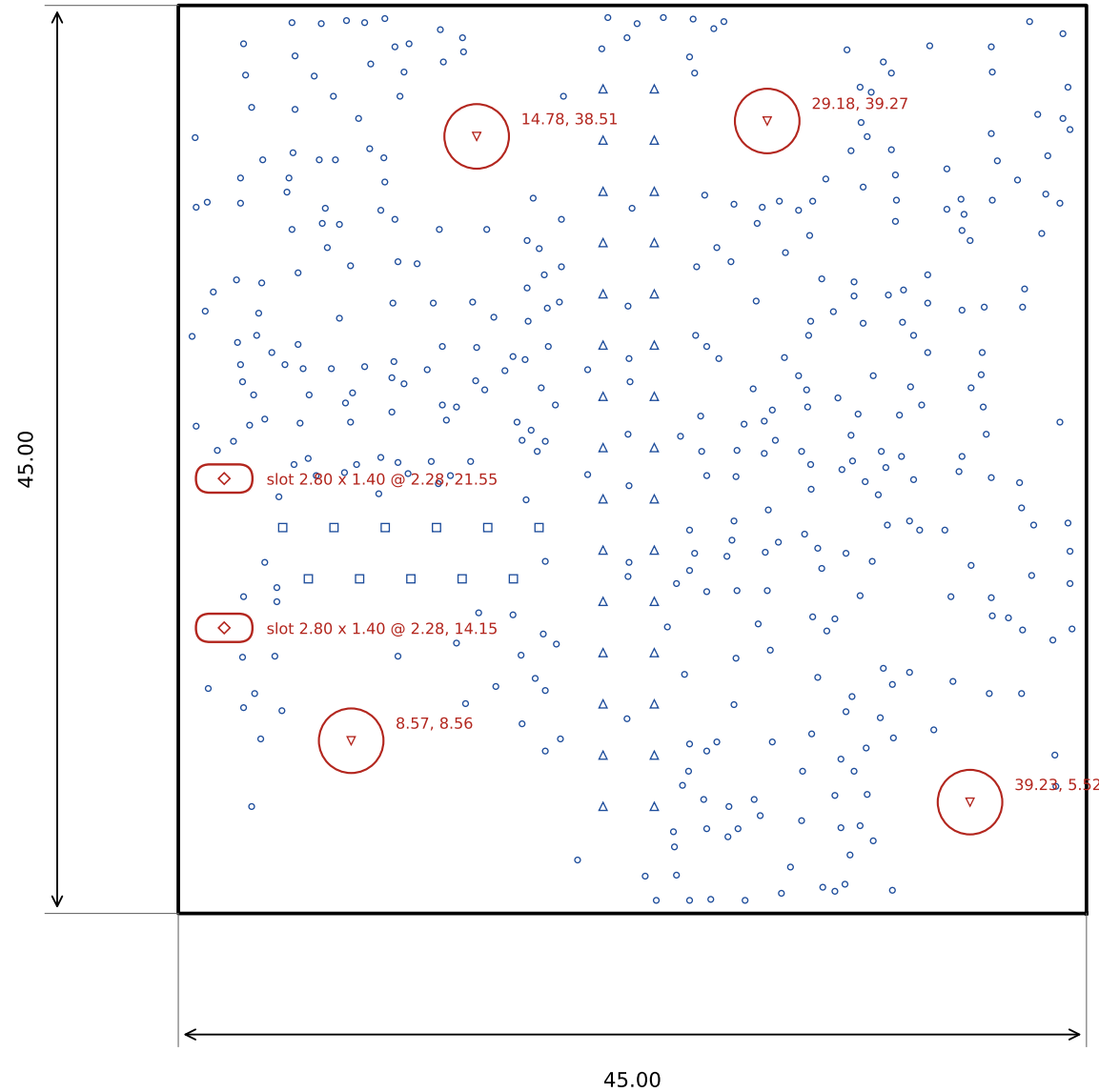
1. **ALL VIAS EPOXY FILLED AND COPPER CAPPED (resin plugged, plated over; IPC-4761 Type VII). 188 of the 393 vias sit in SMD pads. Via drills 0.30 mm (JLCPCB filled-via range 0.2-0.5 mm). Order option "Via Covering: Epoxy Filled & Capped".**
2. Material FR-4, Tg 155 C or better (JLCPCB "FR4 TG155", Nan Ya NP-155F). 8 copper layers. Finished thickness 1.6 mm +/-10 %.
3. Copper 1 oz (35 um) finished on all eight layers: outer 1 oz, inner 1 oz (order option "Inner copper weight 1 oz"; the fab default of 0.5 oz inner is NOT acceptable).
4. Stackup: the fab's standard 8-layer 1.6 mm build (no stackup specified), as in the table. No impedance control.
5. Surface finish ENIG. Solder mask green LPI both sides. Silkscreen white both sides.
6. Minimum track / space: 0.15 / 0.20 mm on every layer. Minimum drill 0.30 mm (vias); smallest plated pad hole 1.00 mm.
7. Minimum via annular ring 0.10 mm; minimum plated pad annular ring 0.33 mm. Via-to-via hole gap >= 0.30 mm; filled via to any pad hole >= 0.59 mm.
8. Plated hole tolerance +0.13 / -0.08 mm. J11 2x15 header holes are drilled 1.05 mm for Samtec 0.64 mm square posts.
9. Non-plated holes: 4 x 3.20 mm (standoffs) and 2 slots 2.80 x 1.40 mm (J9 cable tie), routed; the slots are G85 routes in the NPTH drill file.
10. Outline on the Edge.Cuts (GM1) layer, tolerance +/-0.2 mm.
11. Electrical test: 100 % flying probe; IPC-D-356 netlist BOOST_control_RC1.ipc356 supplied.
12. No order number or other marking to be added ("Remove Mark").
13. Assembly panel: standard PCBA needs at least 70 x 70 mm; order "Panel by JLCPCB" (edge rails) for this 45 x 45 mm card.

STACKUP (from the board file; copper + dielectric 1.654 mm, + mask)

F.Mask	solder mask		0.010
F.Cu	copper, 1 oz	signal / parts	0.035
	prepreg 2116 NP-155F, er 4.16		0.109
In1.Cu	copper, 1 oz	GND plane	0.030
	core NP-155F, er 4.23		0.250
In2.Cu	copper, 1 oz	signal	0.030
	prepreg 2 x 2116		0.218
In3.Cu	copper, 1 oz	signal / 5V pour	0.030
	core NP-155F		0.250
In4.Cu	copper, 1 oz	signal / GND fill	0.030
	prepreg 2 x 2116		0.218
In5.Cu	copper, 1 oz	signal	0.030
	core NP-155F		0.250
In6.Cu	copper, 1 oz	GND plane	0.030
	prepreg 2116		0.109
B.Cu	copper, 1 oz	signal / parts	0.035
B.Mask	solder mask		0.010

DRILL TABLE

sym	drill mm	type	count	used for
○	0.30	PTH	393	vias (filled, capped)
□	1.00	PTH	11	J9 wire pads
△	1.05	PTH	30	2x15 header (J11)
◇	1.40	NPTH slot	2	J9 tie slots 2.8 x 1.4
▽	3.20	NPTH	4	standoffs



Datum: lower-left corner of the outline (0, 0). Dimensions in mm, outline tolerance +/-0.2 mm.
Drawn from the Excellon files: every drilled hole is shown with its symbol; holes >= 2 mm also to scale.